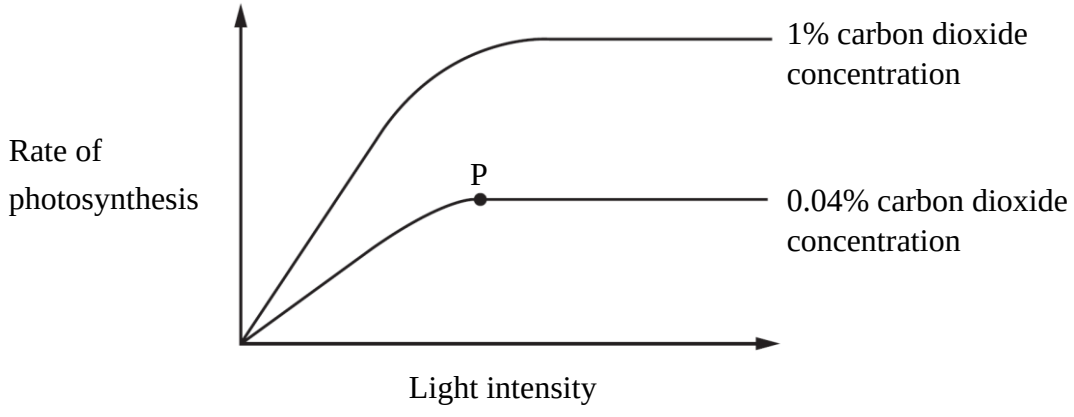
Multiple-choice questions

|!|BCEMB0210000001|!| ★

The graph below shows the changes in the photosynthetic rate of a plant with increasing light intensity at two different carbon dioxide concentrations. The temperature is maintained at 25 °C.



Which of the following is the factor limiting the rate of photosynthesis at point P?

A. light intensity
B. temperature
C. concentration of carbon dioxide
D. concentration of oxygen

□

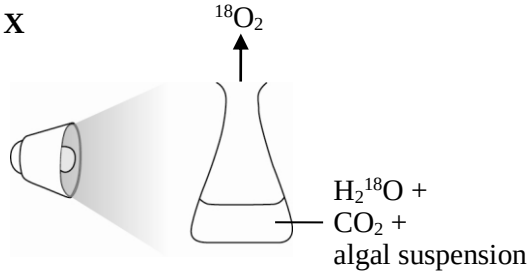
##C##

Skill assessed	Mark
Analysing data	1

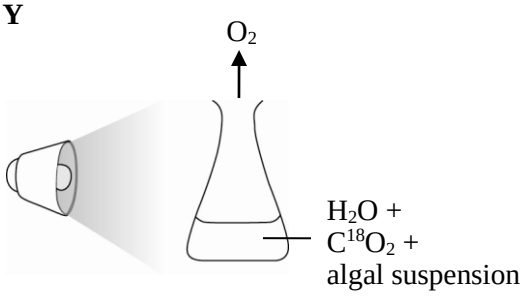
|!|BCEMA0210000002|!| ★★ **SI**

The diagrams below show two experiments, X and Y. In experiment X, an algal suspension was enriched with $H_2{}^{18}O$. In experiment Y, an algal suspension was enriched with $C^{18}O_2$.

Experiment X



Experiment Y



According to the results shown, which of the following can be concluded?

(1) The oxygen gas evolved during photosynthesis comes from water.
(2) The carbon in the carbohydrates formed during photosynthesis comes from carbon dioxide.
(3) The oxygen in the carbohydrates formed during photosynthesis comes from water.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

□

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0210000003|!|

The light energy absorbed by chlorophyll is converted to chemical energy in the form of ATP during the process of

A. glycolysis.
B. Calvin cycle.
C. photolysis of water.
D. photophosphorylation.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0210000004|!| ★

Which of the following processes require(s) energy from ATP?

(1) carbon dioxide fixation into a 3-C compound
(2) reduction of the 3-C compound to form triose phosphate
(3) regeneration of the carbon dioxide acceptor

A. (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0210000005|!|

Which of the following statements about photosynthesis is correct?

A. Photosynthesis is a catabolic process.
B. Photosynthesis cannot occur in anaerobic condition.
C. Light energy is converted to the forms of heat and ATP during photosynthesis.
D. Photosynthesis helps maintain the balance between oxygen and carbon dioxide in the atmosphere.

##D##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0210000006|!| ★

Which of the following is ***not*** the significance of photosynthesis?

A. Photosynthesis produces raw materials for the synthesis of lipids and proteins.
B. Photosynthesis consumes water and thus creates a transpiration pull.
C. Photosynthesis allows the fixation of carbon from carbon dioxide in the atmosphere into plant biomass.
D. Photosynthesis maintains the energy flow in the ecosystem.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0210000007|!|

Which of the following substances is/are the reactant(s) of photosynthesis?

(1) water
(2) carbon dioxide
(3) oxygen

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0210000008|!| **SI**

Directions: The following two questions refer to the steps listed below that test for starch in a green leaf.

(1) Add a few drops of iodine solution to the leaf.
(2) Immerse the leaf in hot alcohol.
(3) Soften the leaf in hot water.
(4) Immerse the leaf in boiling water for one minute.

The correct sequence of the steps is

A. (4) → (3) → (2) → (1).
B. (2) → (3) → (1) → (4).
C. (3) → (2) → (4) → (1).
D. (4) → (2) → (3) → (1).

##D##

Skill assessed	Mark
Practical skills	1

|!|BCEMC0210000009|!|

In procedure (2), it is advised to heat the alcohol in a hot water bath instead of heating the alcohol directly. This can prevent

A. the leaf from breaking into pieces.
B. cracking the test tube.
C. the alcohol from catching fire.
D. damage to the chlorophyll in the leaf.

##C##

Skill assessed	Mark
Practical skills	1

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|!|BCEMC0210000010|!|

A fresh *Hydrilla* carries out photosynthesis and releases gas bubbles. Which of the following combinations correctly shows the identity of the gas and a suitable test to identify the gas?

	Gas	***Test***
A.	hydrogen	burns with a ‘pop’ sound
B.	hydrogen	relights a glowing splint
C.	oxygen	burns with a ‘pop’ sound
D.	oxygen	relights a glowing splint

##D##

Skill assessed	Mark
Practical skills	1

|!|BCEMC0210000011|!|

Which of the following is ***not*** necessary for photosynthesis to take place in green plants?

A. water
B. chlorophyll
C. carbon dioxide
D. oxygen

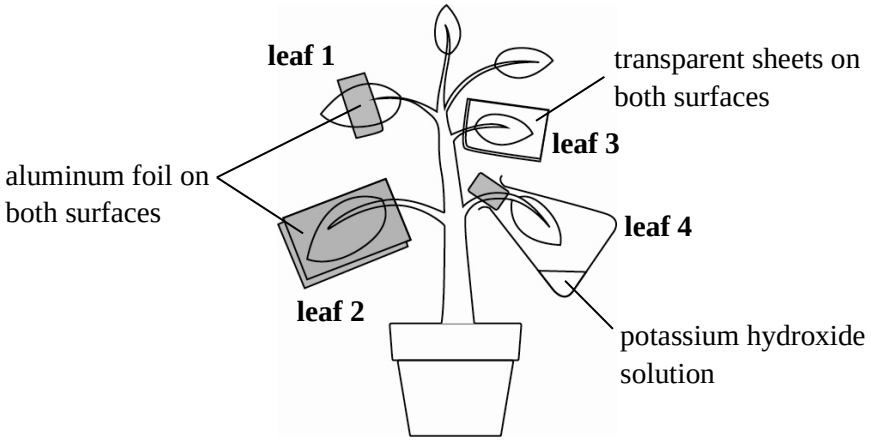
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMA0210000012|!| ★★ **SI**

Directions: The following three questions refer to the diagram below which shows an experiment to study the requirements for photosynthesis. A destarched plant had been exposed to sunlight for four hours before its leaves were tested for starch.



Before the experiment, the plant should be destarched by

A. leaving the plant unwatered for two days.
B. smearing the leaves with vaseline.
C. keeping the plant in a dark room for two days.
D. putting the plant inside a bell jar.

☐

##C##

Skill assessed	Mark
Practical skills	1

|!|BCEMA0210000013|!| ★★ **SI**

Referring to the above experiment, which of the leaves would turn brown when tested with iodine solution?

A. leaf 1 and leaf 2
B. leaf 2 and leaf 3
C. leaf 2 and leaf 4
D. leaf 3 and leaf 4

☐

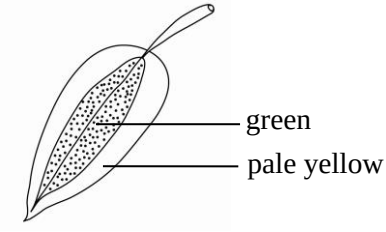
##C##

Skill assessed	Mark
Analysing data	1

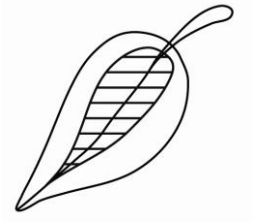
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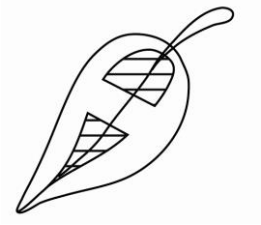
|!|BCEMA0210000014|!| ★★ **SI**

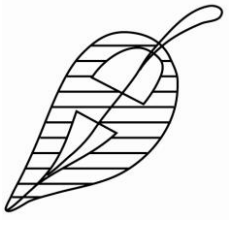
The above experiment was repeated by using a potted plant with variegated leaves. One of its leaves is shown below.

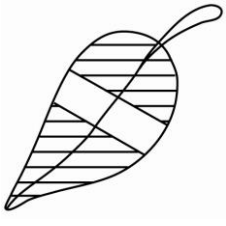


Which of the following correctly shows the result of iodine test on leaf 1 (variegated leaf)?

A. 

B. 

C. 

D. 

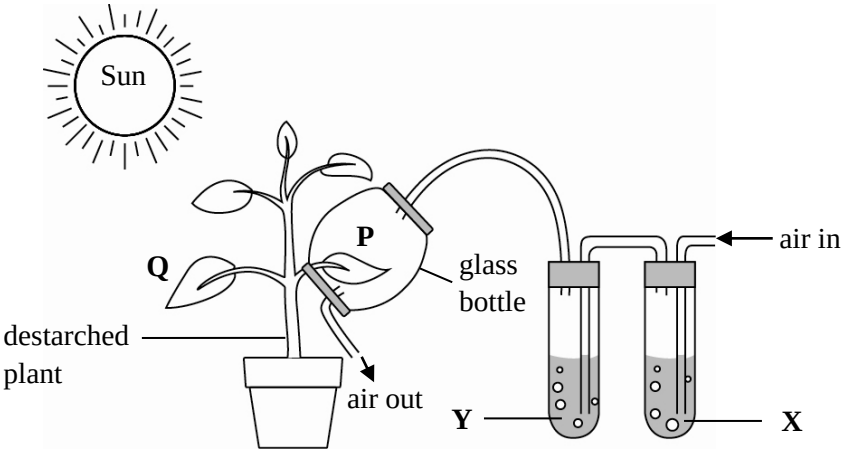
Key: ☐ = brown ▤ = blue black

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0210000015|!| ★ SI

Directions: The following two questions refer to the experimental set-up below which is used to demonstrate the need of carbon dioxide for photosynthesis. The set-up was left exposed to strong sunlight for three hours, and then leaf P and leaf Q were tested with iodine solution.



What are liquid X and liquid Y?

	Liquid X	***Liquid Y***
A.	potassium hydroxide solution	lime water
B.	potassium hydroxide solution	sodium hydrogencarbonate solution
C.	lime water	potassium hydroxide solution
D.	lime water	iodine solution

□

##A##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0210000016|!| ★ SI

At the end of the above experiment, what would be the colours of liquid X, liquid Y, leaf P and leaf Q?

	Liquid X	***Liquid Y***	***Leaf P***	***Leaf Q***
A.	colourless	milky	brown	blue black
B.	milky	colourless	blue black	brown
C.	milky	milky	blue black	brown
D.	colourless	colourless	brown	blue black

□

##D##

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Skill assessed	Mark
Analysing data	1

|!|BCEMB0210000017|!| ★

Which colours of light does chlorophyll absorb the most and the least?

	Absorbs the most	*Absorbs the least*
A.	red	blue
B.	blue	red
C.	red	green
D.	green	red

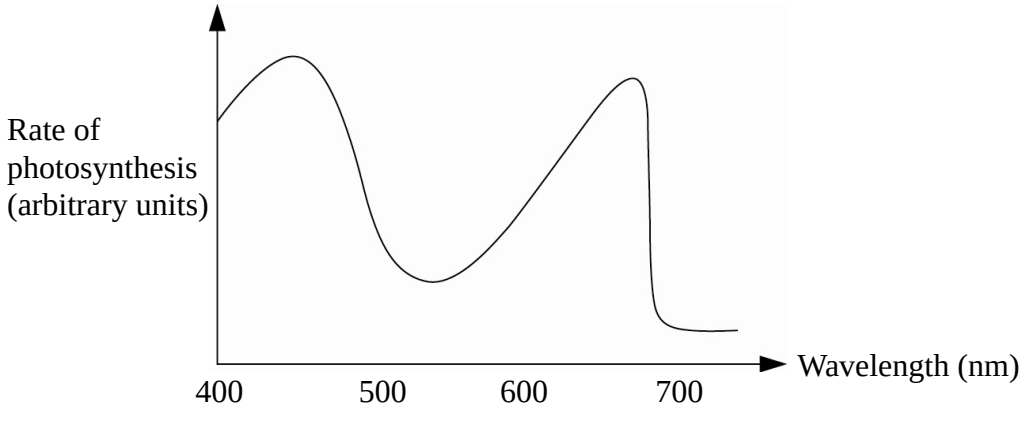
□

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0210000018|!| ★

Directions: The following two questions refer to the graph below which shows the rate of photosynthesis of green plants in different wavelengths of light, and the table which shows the wavelength range of different colours of light.



Colour	Wavelength (nm)
Violet	380–450
Blue	450–485
Cyan	485–500
Green	500–565
Yellow	565–590
Orange	590–625
Red	625–740

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Which colours of light are the most effective for photosynthesis in green plants?

A. violet and blue
B. blue and red
C. violet and green
D. green and red

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0210000019|!| ★

If only green light could reach the Earth, which of the following would happen?

A. The amount of carbon dioxide in the air would decrease.
B. The amount of food available for other organisms would decrease.
C. The amount of oxygen in the air would increase.
D. The average temperature of the Earth would decrease.

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0210000020|!| ★ **STSE**

Which of the following statements about how the conditions in greenhouses lead to an increased crop yield is ***incorrect***?

A. An optimal light intensity can be provided regardless of the daylight hours.
B. Carbon dioxide generators are installed to keep the carbon dioxide concentration higher than the atmospheric level.
C. A cool temperature is maintained to prevent the crops from overheating.
D. The transparent roof and walls of a greenhouse let sunlight in and trap heat inside.

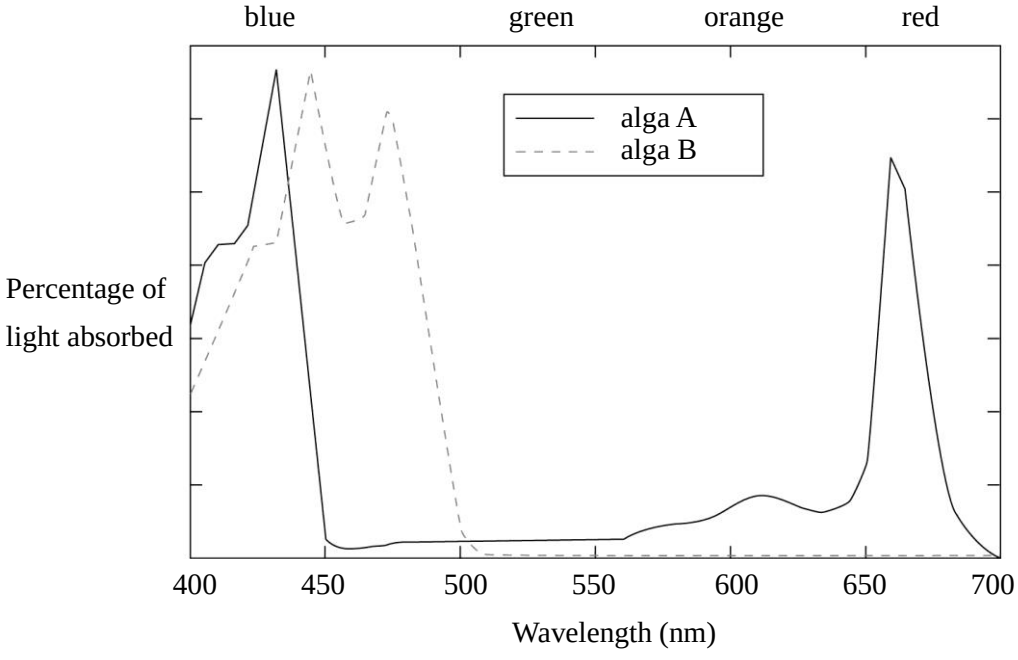
##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0210000021|!| ★

The graph below shows the percentage of light of different colours absorbed by two types of algae, A and B.



A water sample containing algae A and B was collected. Which colour of light can be used to differentiate alga A from alga B by promoting the growth of alga A but inhibiting that of alga B?

A. red
B. green
C. blue
D. purple

□

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0210000022|!|

Which of the following is ***not*** an adaptation of a leaf for photosynthesis?

A. Palisade mesophyll cells contain numerous chloroplasts.
B. The veins contain vascular bundles.
C. Many air spaces are found among the spongy mesophyll cells.
D. Stomata close when the plant is directly lit by sunlight.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0210000023|!|

In dicotyledonous plants, there are usually more palisade mesophyll cells situated in the upper part of the leaves than in the lower part. Which of the following is the best explanation for this?

A. More light can be absorbed by the palisade mesophyll cells.
B. Less water will be lost from the palisade mesophyll cells.
C. Gas exchange is inefficient in the lower part of the leaves.
D. Sugars can be transported away from the palisade mesophyll cells more quickly.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0210000024|!|

Which of the following is ***not*** a function of guard cells?

A. They regulate the amount of carbon dioxide passing through the stomata.
B. They allow more light to penetrate into the photosynthetic cells.
C. They can carry out photosynthesis to synthesize carbohydrates.
D. They prevent excessive water loss.

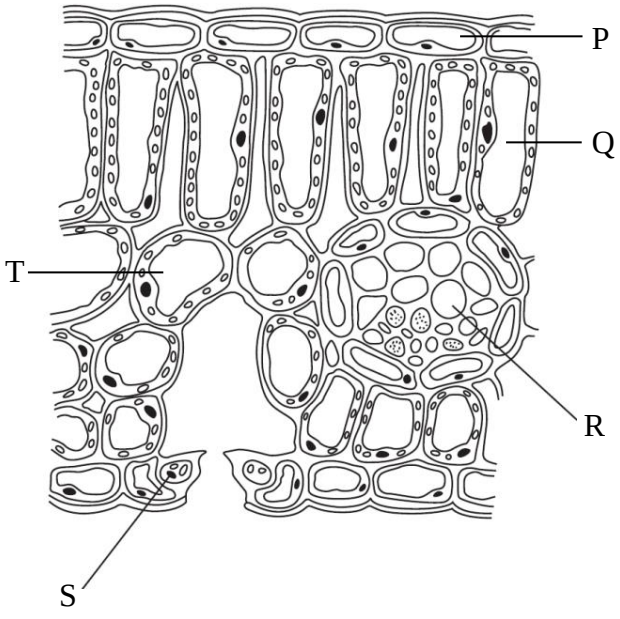
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0210000025|!|

Directions: The following two questions refer to the diagram below which shows a transverse section of a leaf.



Which of the labelled cells has the highest rate of photosynthesis?

A. P
B. Q
C. S
D. T

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0210000026|!| **Cross topic**

Which of the following is ***not*** a feature of cell R in the above diagram?

A. It does not contain nucleus.
B. It transports the carbohydrates produced by photosynthetic cells away from the leaf.
C. It has lignified cell walls.
D. It does not contain cytolpasm.

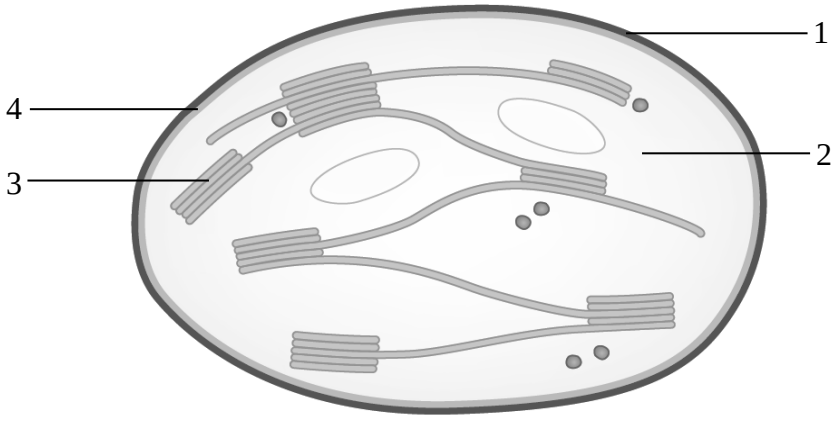
##B##

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Skill assessed	Mark
Analysing data	1

|!|BCEMB0210000027|!| ★

The diagram below shows the structure of a chloroplast.



Which of the following is the site where photolysis occurs?

A. 1
B. 2
C. 3
D. 4

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0210000028|!|

Which of the following statements about the stages of photosynthesis is ***incorrect***?

A. In photochemical reactions, water molecules are split into oxygen and hydrogen.
B. In photochemical reactions, ATP is broken down to form ADP.
C. Light is not required in the Calvin cycle.
D. A leaf takes in atmospheric carbon dioxide as a source of carbon in the Calvin cycle.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0210000029|!| ★

Which of the following substances are produced in the photochemical reactions of photosynthesis?

(1) ADP
(2) NADPH
(3) oxygen
(4) triose phosphate

A. (1) and (2) only
B. (2) and (3) only
C. (1), (2) and (3) only
D. (2), (3) and (4) only

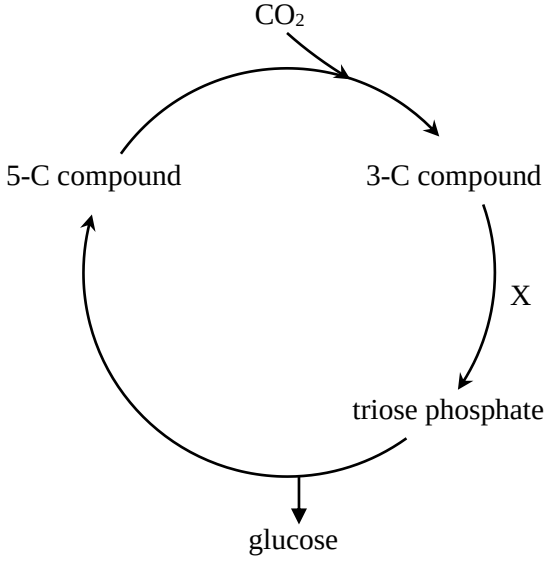
□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0210000030|!| ★

The diagram below summarizes the major steps of the Calvin cycle.



What of the following is/are needed to reduce the 3-C compound to triose phosphate in stage X?

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(1) ATP
(2) NADH
(3) NADP

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0210000031|!|

Where does the Calvin cycle take place in a plant cell?

A. the stroma of a chloroplast
B. the inner membrane of mitochondrion
C. the thylakoid of a chloroplast
D. the cytoplasm

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0210000032|!|

Which of the following are produced in the Calvin cycle?

(1) ADP
(2) triose phosphate
(3) NADPH

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0210000033|!| ★

In the Calvin cycle, the synthesis of triose phosphate is important because it can be used to

(1) form glucose.
(2) regenerate the 5-C carbon dioxide acceptor.
(3) regenerate NADP.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0210000034|!|

Which of the following processes occurs in the Calvin cycle?

A. synthesis of ATP through photophosphorylation
B. splitting of water molecules into hydrogen and oxygen
C. synthesis of NADPH
D. synthesis of triose phosphate

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0210000035|!| ★

In photosynthesis, the Calvin cycle is essential for the photochemical reactions to occur. This is because in the Calvin cycle,

(1) NADP is generated.
(2) ATP is generated.
(3) the 5-C carbon dioxide acceptor is regenerated.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0210000036|!| ★

Which of the following is the fate of triose phosphate formed during the Calvin cycle?

A. It will combine with carbon dioxide to form a 3-C compound.
B. It will be used to regenerate the 5-C carbon dioxide acceptor.
C. It will accept hydrogen and form NADPH.
D. It will be reduced to form a 3-C compound.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0210000037|!|

How many molecule(s) of carbon dioxide is/are needed to be fixed in order to make one molecule of glucose?

A. 1
B. 3
C. 6
D. 12

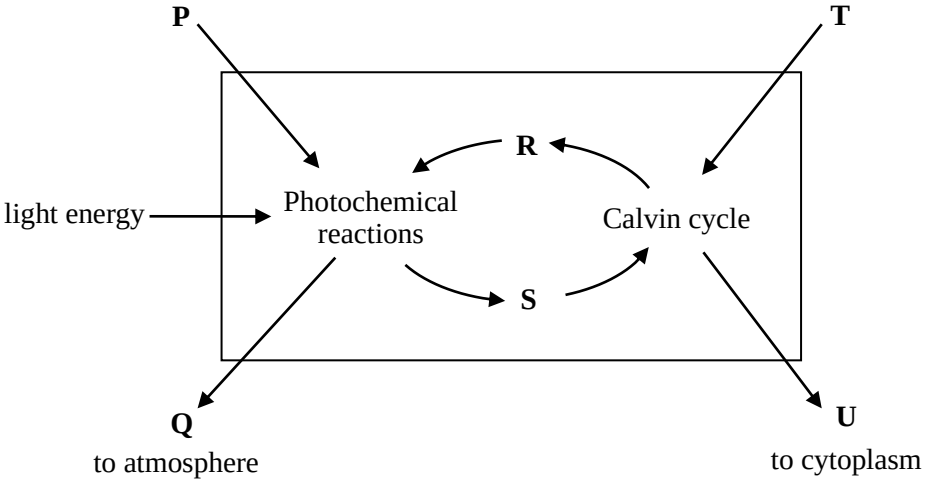
##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0210000038|!| ★

Directions: The following three questions refer to the diagram below which summarizes the process of photosynthesis.



The rectangle represents the site in a cell where photosynthesis occurs.
P to U are substances involved in photosynthesis.

Substances P and T are, respectively,

A. chlorophyll and water.
B. water and carbon dioxide.
C. water and oxygen.
D. oxygen and carbon dioxide.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0210000039|!| ★

Substances Q and U are, respectively,

A. oxygen and glucose.
B. carbon dioxide and glucose.
C. carbon dioxide and water.
D. oxygen and water.

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0210000040|!| ★

Which of the following are the possible identities of substances R and S?

	R	***S***
A.	ADP	NADP
B.	ATP	NADPH
C.	NADP	ATP
D.	NADPH	ADP

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0210000041|!| ★★ SI

A photosynthesizing plant was provided with ^{14}C-labelled carbon dioxide. After a certain period of time, ^{14}C was found in carbohydrates in the leaves of the plant. This experiment shows that

A. carbon dioxide is necessary for photosynthesis.
B. carbohydrates are formed during photosynthesis.
C. plants need ^{14}C-labelled carbon dioxide to produce carbohydrates during photosynthesis.
D. the carbon in the carbohydrates produced during photosynthesis comes from carbon dioxide.

□

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0210000042|!|

The following flowchart shows the different stages in the synthesis of proteins in plants.



Which of the following stages require(s) nitrate ions?

A. stage Y only
B. stage Z only
C. stages X and Y only
D. stages Y and Z only

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0210000043|!|

Starch in leaves can be converted to substance M which can be transported away from the leaves through structure N. What are M and N respectively?

	M	***N***
A.	sucrose	xylem
B.	glucose	phloem
C.	sucrose	phloem
D.	glucose	xylem

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0210000044|!| ★

Which of the following are the functions of photosynthetic products in plants?

(1) They can be used to release energy through respiration.
(2) They can be converted to starch for storage.
(3) They can be used to synthesize proteins for making new cells.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0210000045|!|

Which of the following correctly shows the usual form of carbohydrates found in the phloem and in the storage organ of plants?

	Phloem	*Storage organ*
A.	sucrose	starch
B.	glucose	starch
C.	sucrose	sucrose
D.	glucose	sucrose

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0210000046|!| ★

Which of the following correctly shows the way by which carbohydrates are stored in animals and plants?

	In animals	*In plants*
A.	stored as glycogen in liver cells	stored as starch in the root cortex
B.	stored as glycogen in liver cells	stored as sucrose in the stem phloem
C.	stored as lipid in muscle cells	stored as starch in the root cortex
D.	stored as lipid in muscle cells	stored as sucrose in the stem phloem

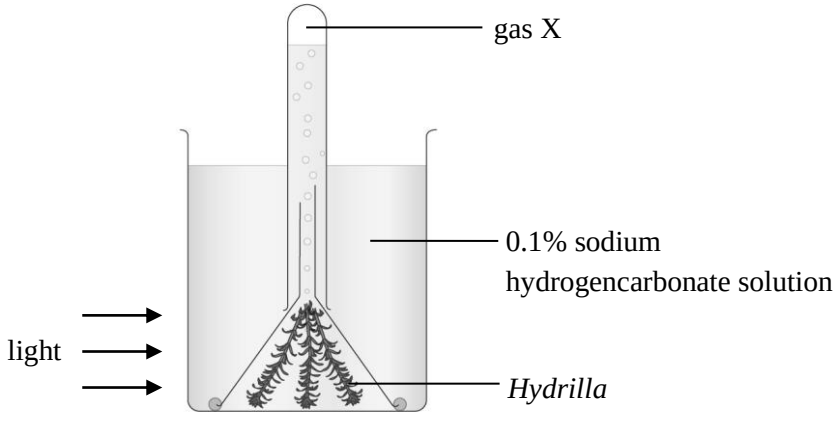
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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0210000047|!| ★ SI

Directions: The following two questions refer to the diagram below which shows an experimental set-up to investigate the effect of carbon dioxide concentration on the rate of photosynthesis.



Which of the following is/are the independent variable(s) under investigation?

(1) light intensity
(2) concentration of sodium hydrogencarbonate solution
(3) volume of gas X collected

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

□

##B##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0210000048|!| ★ SI

Which of the following combinations correctly shows a suitable test to identify gas X and the role of gas X in photosynthesis?

	Test for gas X	*Role of gas X*
A.	pass gas X through lime water	reactant
B.	put a glowing splint over the mouth of a test tube containing gas X	reactant
C.	pass gas X through lime water	product
D.	put a glowing splint over the mouth of a test tube containing gas X	product

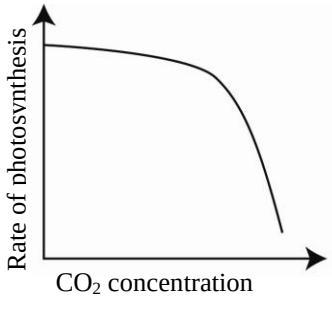
##D##

Skill assessed	Mark
Practical skills	1

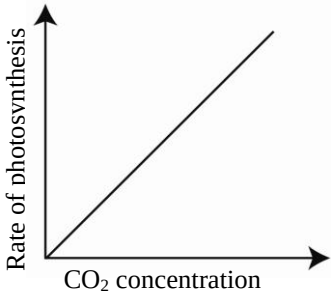
|!|BCEMB0210000049|!| ★

Which of the following graphs correctly shows the effect of carbon dioxide concentration on the rate of photosynthesis?

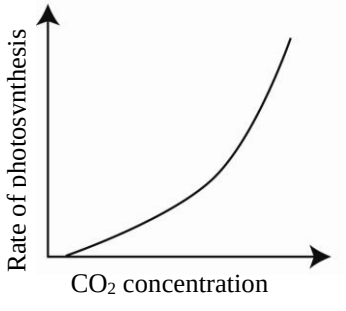
A.



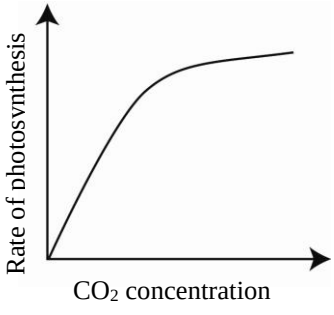
B.



C.



D.

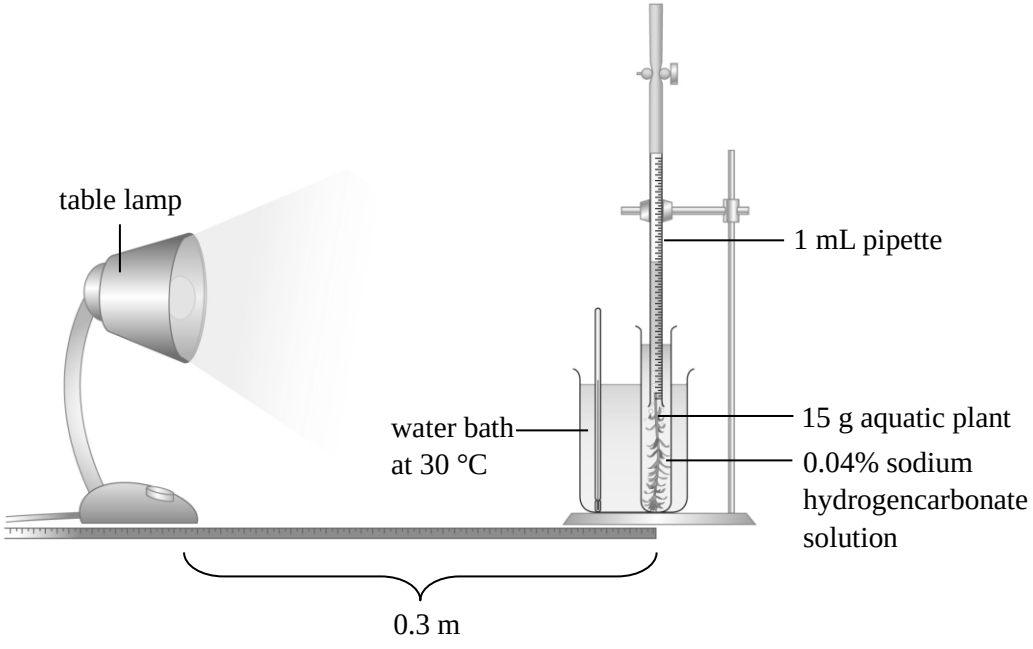


##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0210000050|!| ★ SI

The diagram below shows an experimental set-up used to find out the rate of photosynthesis of an aquatic plant.



After 30 minutes, 0.15 mL of oxygen (O_2) was collected. What is the photosynthetic rate per unit mass of this aquatic plant?

A. 0.01 mL O_2 g^{-1} h^{-1}
B. 0.02 mL O_2 g^{-1} h^{-1}
C. 0.15 mL O_2 g^{-1} h^{-1}
D. 0.3 mL O_2 g^{-1} h^{-1}

□

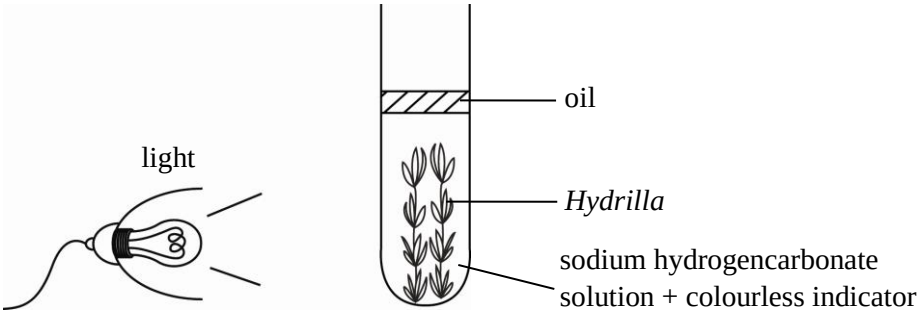
##B##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMB0210000051|!| ★ SI

Directions: The following three questions refer to the diagram below which shows a set-up used to study the rate of photosynthesis of *Hydrilla*. The *Hydrilla* was put into a test tube containing sodium hydrogencarbonate solution and a colourless indicator. The indicator turns red when the level of oxygen is high.



What is the purpose of immersing *Hydrilla* in sodium hydrogencarbonate solution?

A. It provides an optimum pH for the growth of the plant.
B. It provides carbon dioxide for photosynthesis to take place in the plant.
C. It acts as hydrogencarbonate indicator to detect the presence of carbon dioxide.
D. It provides sodium ions for the growth of the plant.

##B##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0210000052|!| ★ SI

What is the significance of adding a layer of oil on top of the solution?

A. It prevents the entry of bacteria and fungi.
B. It prevents atmospheric oxygen from dissolving into the solution.
C. It prevents the escape of oxygen into the atmosphere.
D. It prevents the evaporation of water.

##B##

Skill assessed	Mark
Practical skills	1

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|!|BCEMB0210000053|!| ★ SI

Which of the following is/are appropriate method(s) to measure the dependent variable of the above experiment?

(1) measure the time for indicator to turn red
(2) measure the number of bubbles released by the *Hydrilla* in five minutes
(3) measure the weight of the *Hydrilla*

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

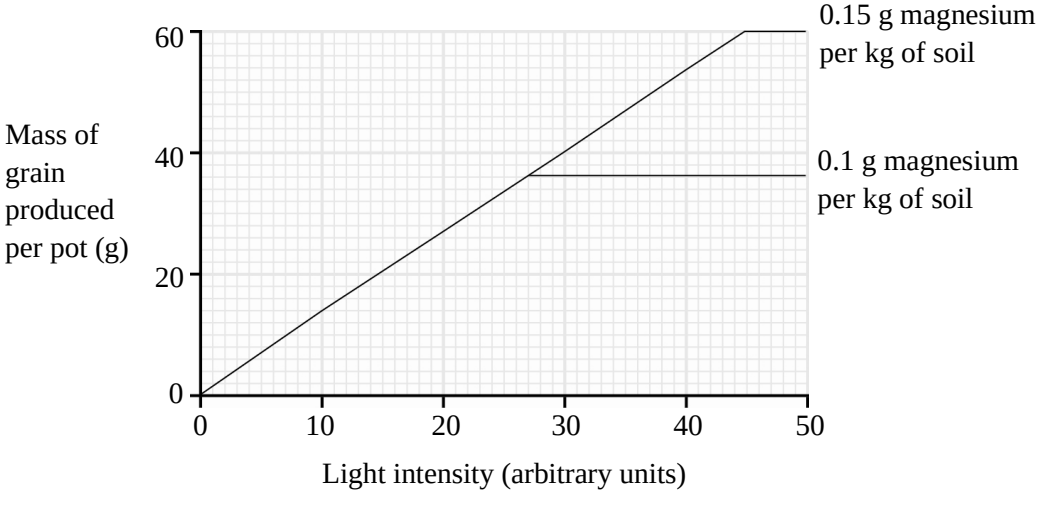
□

##B##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0210000054|!| ★ SI

Diana investigated the effect of magnesium ion concentration on the grain production by a crop plant cultivated at different light intensities. She first prepared 12 pots of soil, the soil either contained 0.1 g or 0.15 g magnesium per kg of soil. She planted 15 crop seeds in each pot and each pot was kept at a different light intensity. After six months, the grains produced by the crop plants were collected and weighed. The following graph shows the results.



According to the graph, which of the following factors limits the grain yield at a light intensity of 20 arbitrary units?

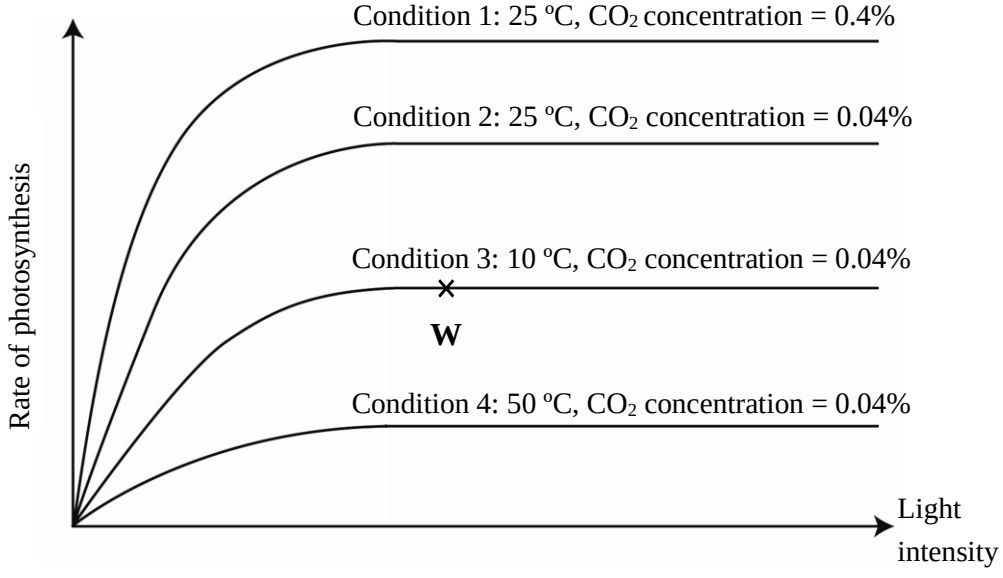
A. light intensity
B. temperature
C. carbon dioxide concentration
D. magnesium ion concentration

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0210000055|!| ★★

Directions: The following three questions refer to an experiment which investigates the influence of light intensity on the rate of photosynthesis of a cotton plant under different conditions. The results are shown in the graph below.



Which of the following can represent the rate of photosynthesis?

A. the rate of oxygen production per unit mass of the plant
B. the rate of water absorption per unit mass of the plant
C. the rate of carbon dioxide production per unit mass of the plant
D. the rate of oxygen consumption per unit mass of the plant

##A##

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Skill assessed	Mark
Practical skills	1

|!|BCEMA0210000056|!| ★★

As shown in the given graph, the rate of photosynthesis under condition 4 is the lowest. Which of the following is the ***best*** explanation?

A. Carbon dioxide concentration becomes the limiting factor of photosynthesis.
B. High temperature increases the rate of transpiration.
C. The concentration of carbon dioxide is low.
D. Enzymes are denatured at 50 °C.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0210000057|!| ★★

Which of the following may be the factor(s) limiting the rate of photosynthesis at point W in the given graph?

(1) light intensity
(2) temperature
(3) carbon dioxide concentration

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

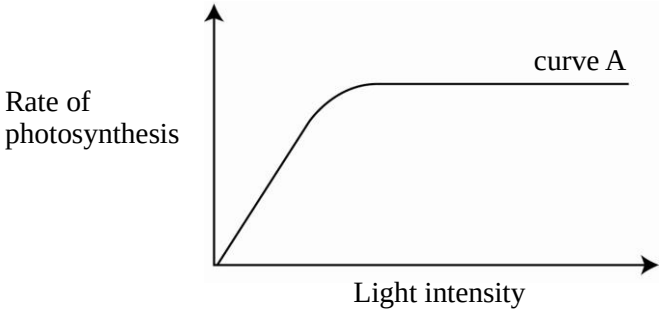
##C##

Skill assessed	Mark
Analysing data	1

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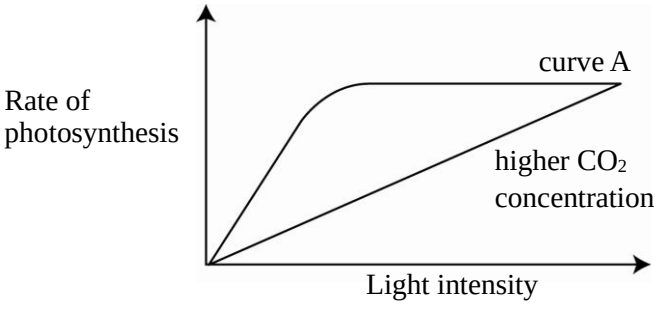
|!|BCEMB0210000058|!| ★

Curve A represents the rate of photosynthesis at a particular carbon dioxide concentration under different light intensity.

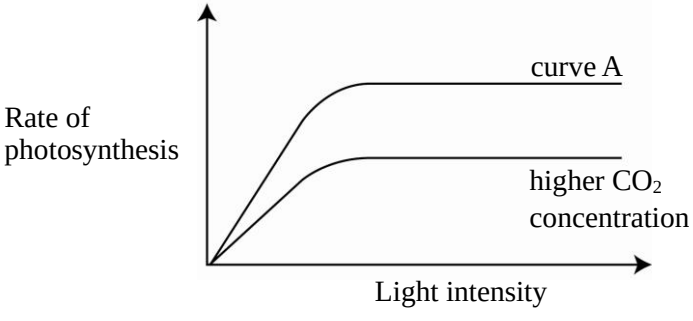


Which of the following graphs ***best*** represents the rate of photosynthesis at a higher carbon dioxide concentration?

A.

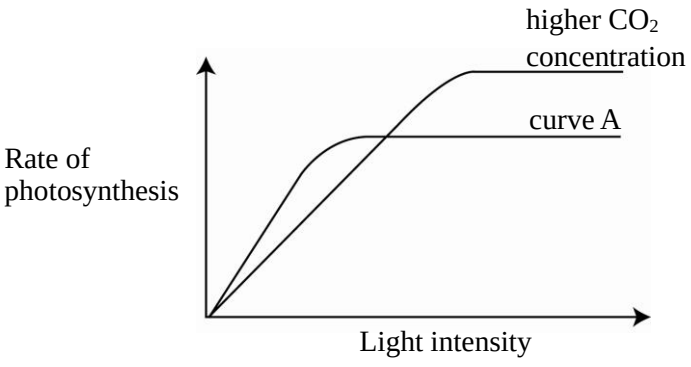


B.

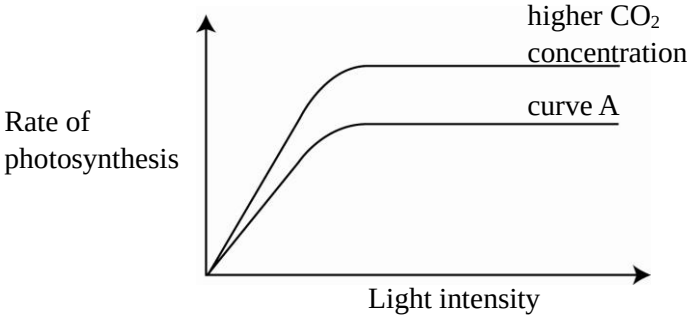


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C.



D.



□

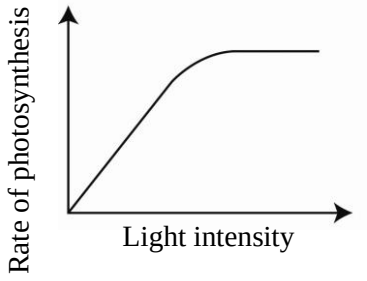
##D##

Skill assessed	Mark
Analysing data	1

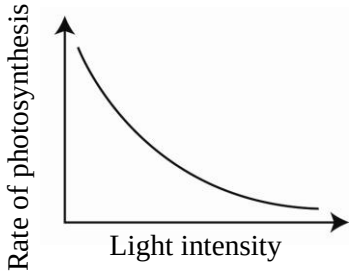
|!|BCEMC0210000059|!|

Which of the following graphs correctly shows the effect of light intensity on the rate of photosynthesis?

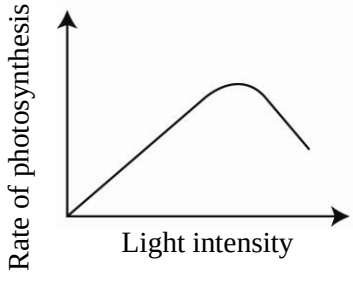
A.



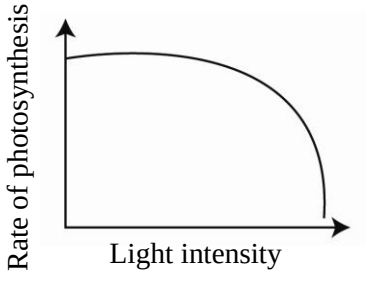
B.



C.



D.



##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0210000060|!| ★

Which of the following factors does ***not*** influence the rate of oxygen production in photosynthesis?

A. temperature
B. light colour
C. number of mitochondria in leaf cells
D. number of chloroplasts in leaf cells

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0210000061|!| ★★

The rate of photosynthesis of desert plants often decreases at midday. Which of the following can ***best*** explain this phenomenon?

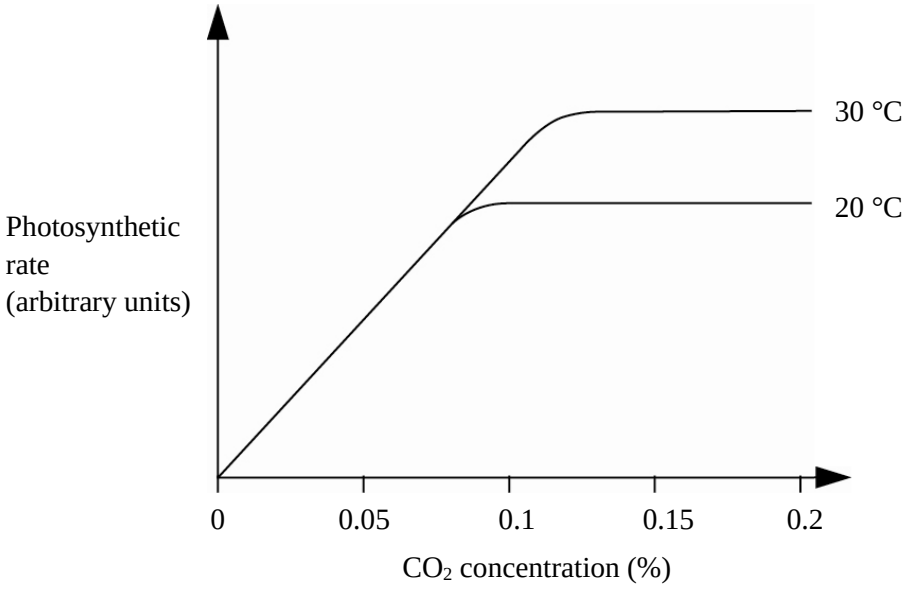
A. There is not enough water for photochemical reactions to occur.
B. Enzymes are denatured at high temperature.
C. Most light is reflected by the waxy cuticle on the leaf.
D. Stomata close to conserve water and so the rate of gas exchange reduces.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0210000062|!| ★

Directions: The following two questions refer to the graph below, which shows the photosynthetic rate of a plant under bright light at 20 °C and 30 °C respectively with different concentrations of carbon dioxide.



Under atmospheric carbon dioxide concentration, what limits the photosynthetic rate of the plant?

A. temperature
B. light intensity
C. water availability
D. carbon dioxide concentration

##D##

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Skill assessed	Mark
Analysing data	1

|!|BCEMB0210000063|!| ★

Which of the following is the most likely explanation for the higher photosynthetic rate at 30 °C than at 20 °C when the carbon dioxide concentration is 0.1%?

A. More carbon dioxide is available for the plant at 30 °C than at 20 °C.
B. The rate of water loss is higher at 30 °C than at 20 °C.
C. The rate of respiration is lower at 30 °C than at 20 °C.
D. Enzyme activity is higher at 30 °C than at 20 °C.

□

##D##

Skill assessed	Mark
Analysing data	1